MILITARY SPECIFICATION

BAND, ROTATING, NYLON GLASS-FILLED FOR PROJECTILE, 105MM, M737 TPDS-T

This specification is approved for use by U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification provides requirements and quality assurance provisions for the fabrication of a rotating band, nylon, glass-filled for use in the 105MM M737 TPDS-T projectile.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition General Specification For

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

FSC 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-1167 - Ammunition Data Card

MIL-STD-1169 - Packaging, Packing and Marking For Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

9335933 - Band, Rotating

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issues listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A322 - Hot Rolled Alloy Steel Bars, Specification for

D785 - Rockwell Hardness of Plastics and Electrical Insulating Materials, Test For

D792 - Specific Gravity and Density of Plastics by Displacement, Test For

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D2290 - Apparent Tensile Strength of Ring or Tubular Plastics By Split Disc Method, Test For

D2584 - Ignition Loss of Cured Reinforced Resins, Test For

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

3. REQUIREMENTS

3.1 General. Materials and parts shall comply with the requirements specified in applicable drawings and specifications.

3.2 Material defects. Soundness of material shall be as required by the applicable specification and drawing. All bands shall be free of blisters, cracks, checks, scratches, voids and any surface or internal defects which may detrimentally affect serviceability or appearance in the judgement of the appropriate technical agency. No more than ten foreign inclusions will be permitted and shall not exceed 1/16 inch in diameter, distributed around the band, not more than four of which may form a cluster within an area of 1/4 inch square.

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Workmanship. Requirements for workmanship are shown on the applicable drawing, and specification plus the following:

3.4.1 Threads. Threads shall be full and undamaged for the entire minimum length or depth and shall be in accordance with the provisions of MIL-A-2550.

3.4.2 Burrs. Bands shall have no burrs which might interfere with the assembly or the performance of the projectile or which might be injurious to personnel handling the item.

3.4.3 Flash. Machined bands must be free of flashing.

3.4.4 Foreign matter. All parts shall be free of chips, dirt, grease, or other foreign matter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions . Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the items in sample quantities as indicated below.

Part Description	Drawing	Quantity
Band, Rotating (machined)	9335933 Sht. 1	25/cavity
Band, Rotating (blank)	9335933 Sht. 2	25/cavity

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-B-63454 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Band, Rotating				9335933
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Blank Band				
	Color (Note 1)			3.1	Visual
	Foreign Matter (Note 1)			3.4.4	Visual
	Density			3.1	4.5.5
	Tensile strength and extension to failure (Note 2)			3.1	4.5.2
	Hardness			3.1	4.5.4
	Glass content (Note 3)			3.1	4.5.8
	Shear strength			3.1	4.5.3
	Machined Band				
	Examination for Defects			3.1	4.4.2.2

NOTES:
1. Perform before any other inspection.
2. Same specimen used in density test.
3. Same specimen used in hardness test.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.1.1 Batch. A batch of raw material shall be defined as a unit of product prepared for shipment from the same lot and may consist of a uniform blend of two or more "production runs" of polyamides of the same composition, type and color.

4.4.1.2 Blank-band lot. A blank band lot shall consist of blanks which have been processed under the same conditions. A blank band lot shall contain only one material batch.

4.4.1.3 Machined-band lot. A machined band lot may be made from one or more blank band lots with the same interfix number.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plan. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63454 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9335933
4.4.2.1	Rotating Band (Blank)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Tensile strength and extension to failure test	See Note 1		3.1	4.5.2
102	Shear strength test	See Note 1		3.1	4.5.3
103	Hardness test	See Note 2		3.1	4.5.4
104	Density test	See Note 2		3.1	4.5.5
105	Moisture content test	See Note 2		3.1	4.5.6
106	Glass content test	See Note 2		3.1	4.5.8
MINOR					
201	Color		0.65%	3.1	Visual
202	Foreign matter		0.65%	3.4.4	Visual
NOTE 1:	After stress relief, ten bands per cavity from each lot (see 4.4.1.2) shall be tested for compliance with the drawing requirements. Failure of two or more samples shall be cause for rejection of the lot. If one sample fails, ten additional blanks per cavity shall be tested. Failure of one or more of these additional samples shall cause rejection of the lot.				

NOTES: 2. After stress relief, ten bands per cavity from each lot (see 4.4.1.2) shall be tested for compliance with drawing requirements. Failure of any sample to meet the applicable requirements shall be cause for rejection of the lot.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63454 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Rotating Band (Machined)				9335933
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101	Diameter, outside		0.40%	3.1	Gage
102	Diameter of counterbore		0.40%	3.1	Gage
103	Pitch diameter of thread, max.		0.40%	3.1	Gage
104	Minor diameter of thread, max.		0.40%	3.1	Gage
105	True position of counterbore with pitch diameter of thread		0.40%	3.1	Gage
106	Runout of outside diameter with pitch diameter of thread		0.40%	3.1	Gage
107	Length from tapered edge to rear basic diameter		0.40%	3.1	Gage
108	Length from tapered edge to forward basic diameter		0.40%	3.1	Gage
109	Profile of taper		0.40%	3.1	Gage
110	Material defects		100%	3.2	4.5.7
MINOR					
201	Length, total		0.65%	3.1	Gage
202	Depth of counterbore		0.65%	3.1	Gage
203	Chamfers missing or incorrect		0.65%	3.1	Visual
204	Pitch diameter of thread, min.		0.65%	3.1	Gage
205	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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4.4.3 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See MIL-A-48078 Section 6 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Testing conditions. Unless otherwise specified in the test method or the applicable specification, the atmospheric conditions surrounding the specimen 24 hours prior to test and during tests shall be 23°C ± 2°C (73.4°F ± 3.6°F) and 50% ± 5% relative humidity. All test specimens are to remain sealed in their plastic shipping bags until testing. The test specimens shall conform as closely as possible to the dimensions shown in Figure 1.

4.5.2 Tensile strength and extension to failure test. Each test specimen conforming to Figure 1 detail 3 shall be tested for tensile strength and extension to failure. The test shall be performed following the criteria of ASTM D2290, procedure B with the following exceptions:

a. The specimen will be uniform in cross section (without reduced section) as detailed in Figure 1.

b. The split-disc test fixture and accessories shall be double pinned and conform to Figure 3 or its approved equivalent. (Ref. 4.3.2)

c. The cross-head speed shall be 0.5 + 0.1 inches per minute.

d. The split discs shall be separated until each half is in contact with the specimen without applying a load to the specimen. The distance of separation (initial separation) shall then be measured. Split disc separation upon specimen rupture (ultimate separation) shall also be measured. Inches of extension shall be determined by subtracting initial separation from ultimate separation. The tensile strength (psi) shall be recorded.

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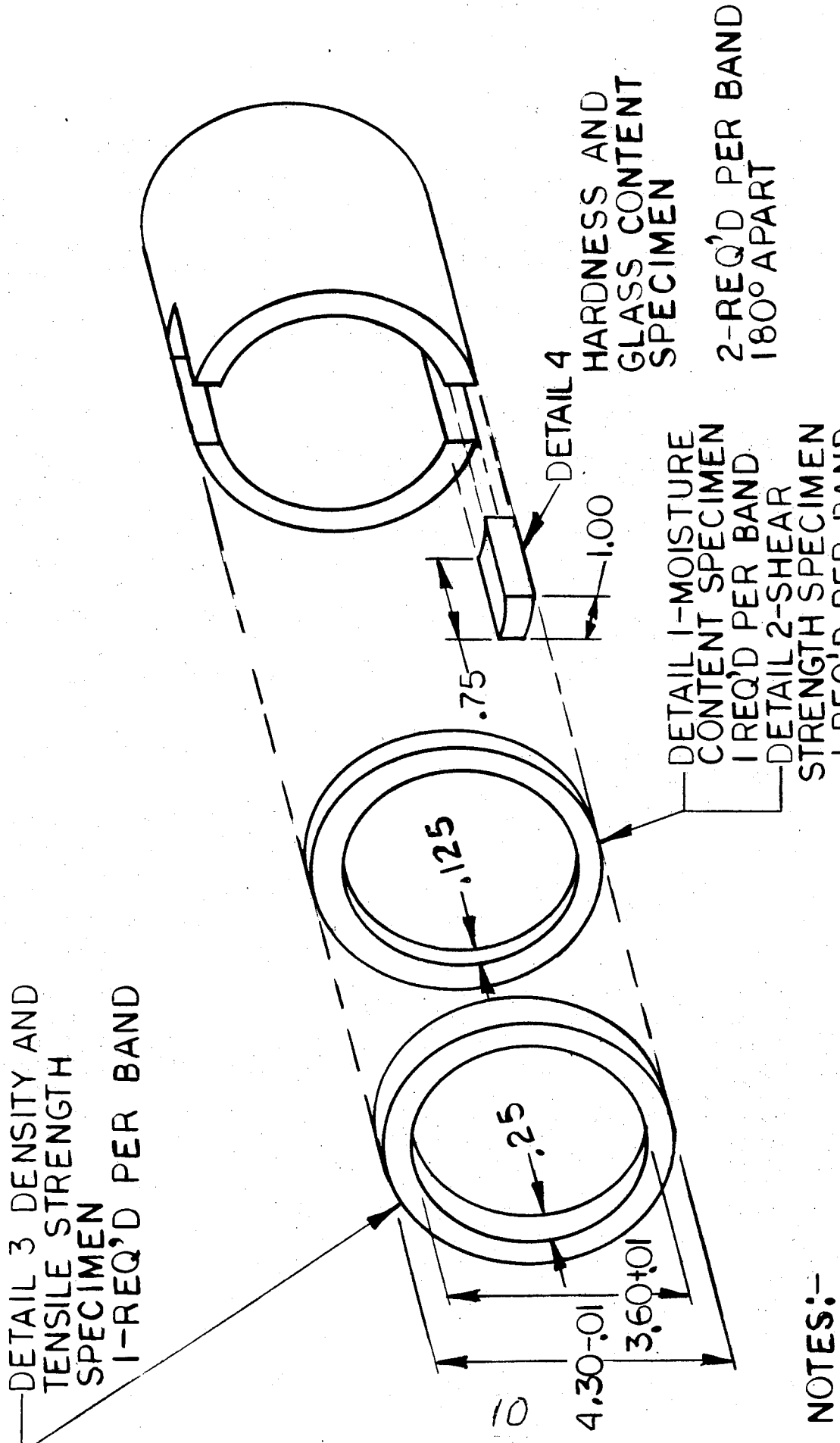
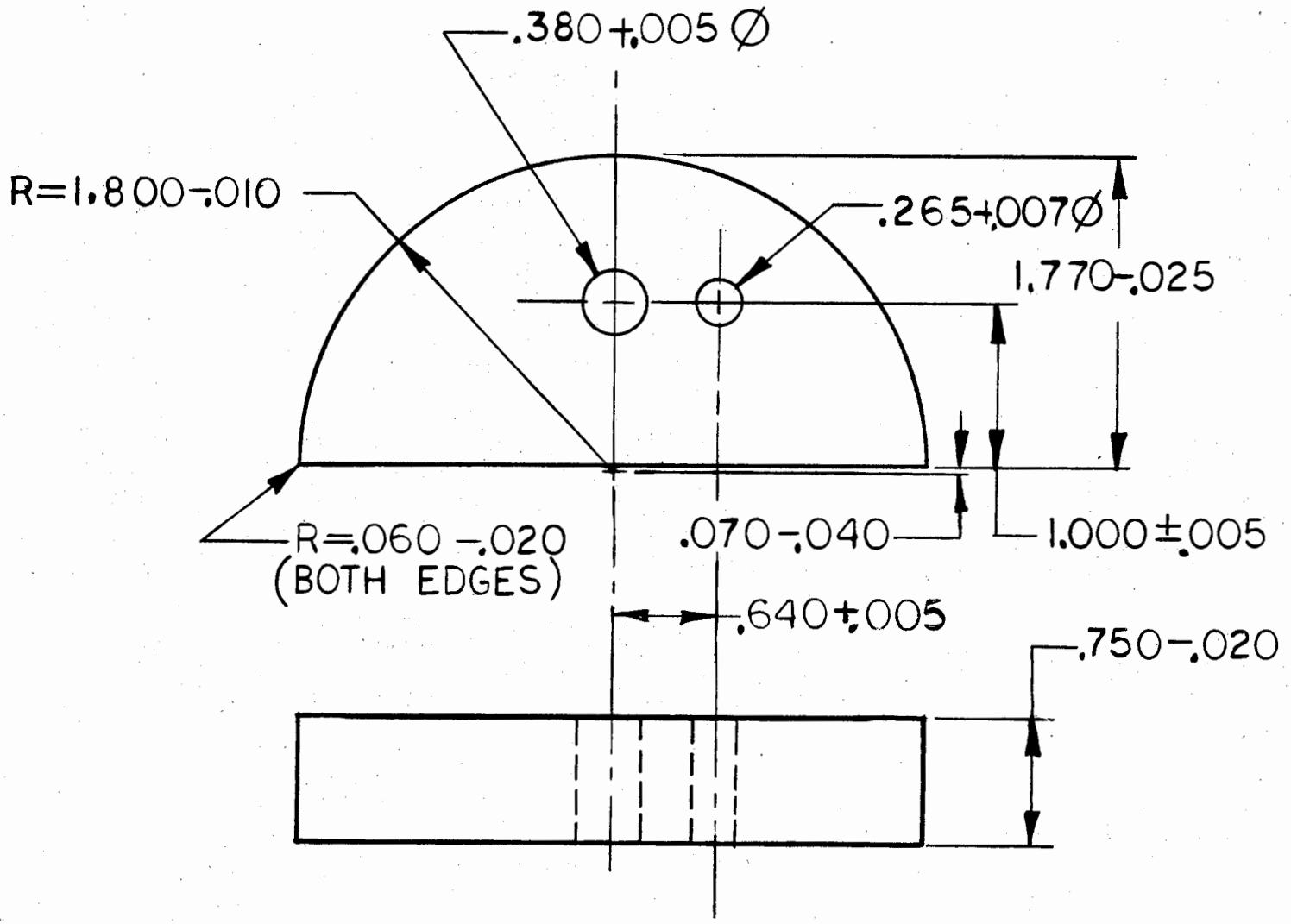


FIGURE 1. NYLON ROTATING BAND TEST SPECIMENS

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MATERIAL-TOOL STEEL

FIGURE 3-SPLIT DISK
TENSILE STRENGTH TEST FIXTURE

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4.5.3 Shear strength test. Each test specimen conforming to of Figure 1 detail 2 shall be placed in the test fixture shown in Figure 2, and tested at a loading rate that will produce failure in 30 to 60 seconds after initial application of the load. The data reported for each specimen shall include actual dimensions, shearing load and shear strength in pounds per square inch. The shear strength shall be determined as follows:

$$\text{Shear Strength (PSI)} = \frac{P}{\pi dt}$$

where:

P = load (lbs.)
d = diameter of punch (inches)
t = thickness of sample (inches)

4.5.4 Hardness test. The inside and outside surfaces of each test specimen conforming to Figure 1 detail 4 shall be tested with a Rockwell Hardness Tester, utilizing the Rockwell "R" scale in accordance with ASTM D785.

4.5.5 Density. Each test specimen conforming to Figure 1 detail 3 shall be tested in accordance with ASTM D792, method A-1. The density in grams per cubic centimeter shall be determined as follows:

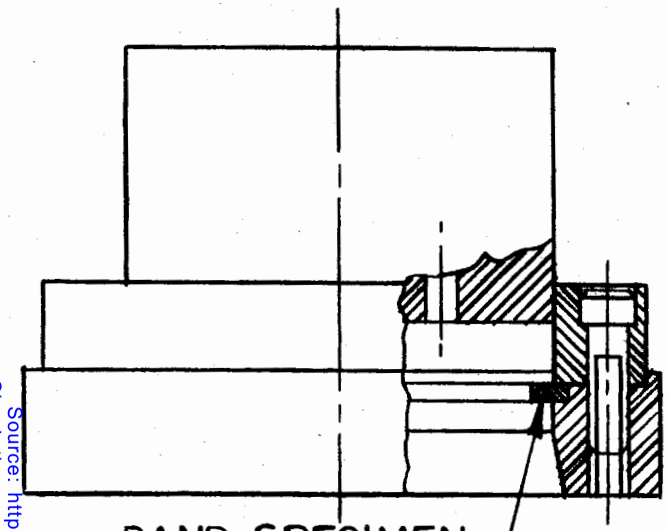
$$\text{Density, (gms./cm}^3) = \frac{D_L \, W_A}{W_A - W_L}$$

Where
D_L = Density of liquid (gms./cm^3)
W_A = Weight of specimen in air, (gms.)
W_L = Weight of specimen in liquid, (gms.)

When water is used "D_L" becomes unity. Care shall be exercised to remove air bubbles surrounding the specimen when being weighed in water.

4.5.6 Moisture content. All specimens will remain sealed in their shipping bags until testing. Remove specimens from their bags, wipe clean of any loose material with a lint-free cloth or tissue, and weigh on an analytical balance. This is the initial weight. For 24 hours, condition the specimens in an oven maintained at 160 ± 2°F (71.1 ± 1.1°C). Then cool them in a desiccator and reweigh. Repeat oven-conditioning, desiccator-cooling, and weighing cycle until the difference between two consecutive weighings is no more than 5 milligrams. This is the conditioned weight. Calculations are as follows:

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NOTES:-

-125 FINISH, EXCEPT AS NOTED.

-MAT'L: STEEL, 4340, SPEC ASTM-A322.

-HEAT TREATMENT RC 45-48.

-REMOVE ALL BURRS

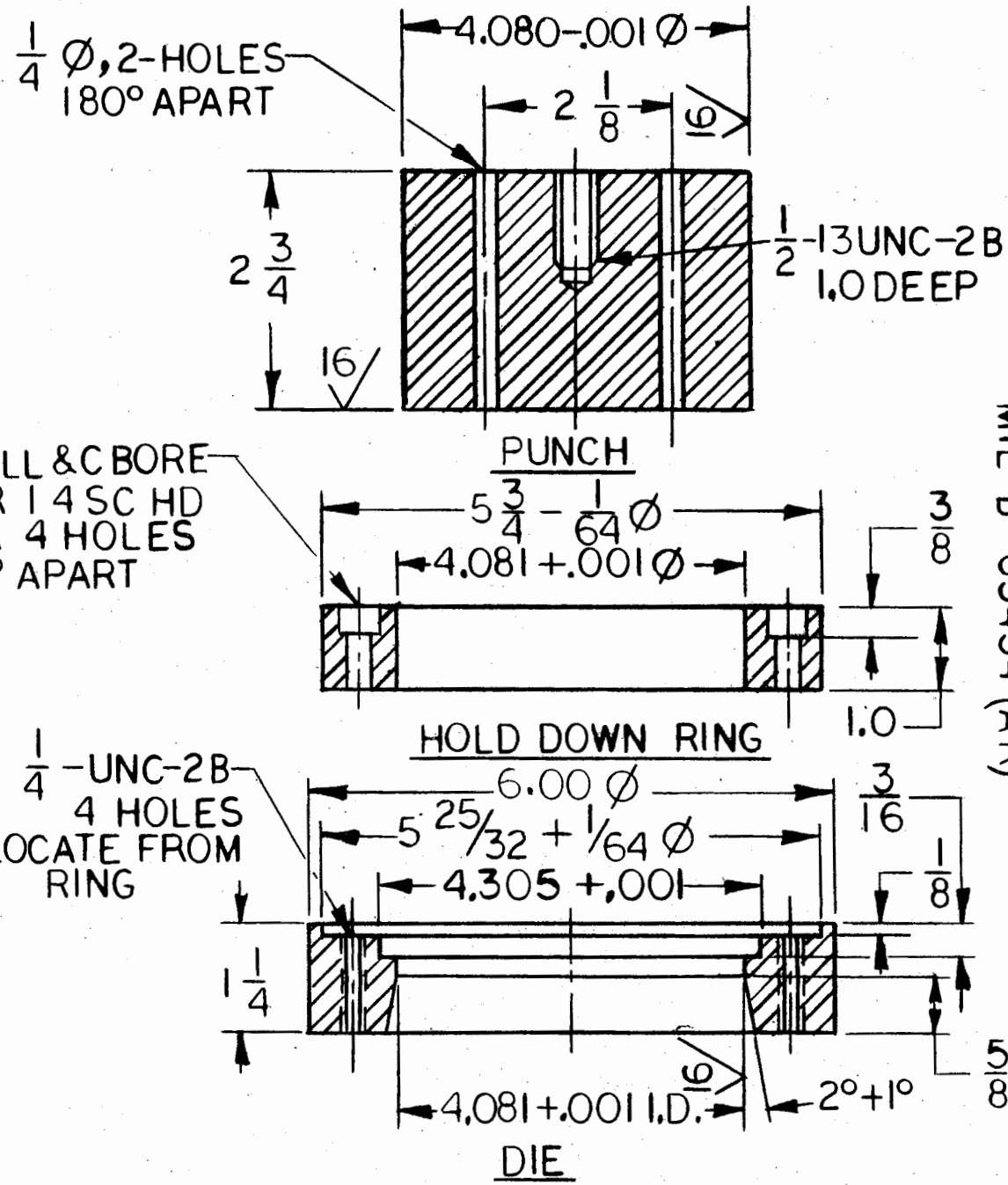


FIGURE 2-SHEAR TEST FIXTURE

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$$\text{Percent Moisture Content} = \frac{\text{Initial weight - Conditioned weight}}{\text{Initial weight}} \times 100$$

4.5.7 Material defects. Internal and external surfaces of bands shall be inspected to determine compliance with 3.2. Each band selected for this inspection shall be illuminated with a minimum of 400 foot-candles. Bands shall be rotated 360° and inspected with an unaided eye. Visual standards shall be established by the contractor for use as references in determining acceptability. The standards shall be approved by the Government prior to use.

4.5.8 Glass content test. Each test specimen conforming to Figure 1 detail 4 shall be tested in accordance with ASTM D2584. The glass content will be calculated by subtracting the ignition loss (%) from 100.

5. PACKAGING

5.1 Packaging requirements. Packaging shall be as specified in MIL-STD-1169 Level B. The packaging used shall protect the bands against moisture absorption.

5.2 Packing and marking. Packing and marking shall be as specified in MIL-STD-1169.

5.4 Shipping. When components from more than one lot are shipped as a car load, each lot must be kept separate and the division between lots clearly indicated to prevent mixing of lots in transit.

6. NOTES

6.1 Visual examination qualification. When compliance with the applicable requirements is in doubt as a result of visual examination, characteristics may be measured or gaged to determine acceptability.

6.2 Ordering data. See MIL-A-48078.

a. Data Cards. Data cards shall be prepared for each lot in accordance with MIL-STD-1167.

b. Delivery of data cards to the technical agency. The following address is to be added to those listed in the contract (Form DD 1423) for the delivery of one copy, to the technical agency of the Ammunition Data Cards:

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Commander, US Army Armament
Research and Development Command
ATTN: DRDAR-QAT-A
Dover, New Jersey 07801

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs, as required to:

Commander
US Army Research and Development Command
ATTN: DRDAR-QAT-I
Dover, NJ 07801

6.4 Intended use. This band will be utilized on Cartridge, 105MM, TPDS-T, M724A1.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1315-A403)

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